

An Explicit Counterexample to the Rank-Two Poisson Conjecture

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July 21, 2026

Abstract

Let

$$\mathcal{P}_2 = \mathbb{C}[x, q, p, z]$$

carry the canonical Poisson bracket determined by $\{p, x\} = \{z, q\} = 1$ and by the vanishing of the other brackets between distinct generators. Here and throughout, “rank two” means two canonical pairs in the standard indexing of the canonical Poisson algebras; thus there are four polynomial generators and the Poisson tensor has geometric rank four. We give explicit polynomials

$$R, T, D, S \in \mathbb{Q}[x, q, p, z]$$

satisfying

$$\{D, R\} = 1, \quad \{S, T\} = 1, \quad \{R, S\} = \{R, T\} = \{D, S\} = \{D, T\} = 0,$$

while

$$R = x(2 - 3xq).$$

Consequently, the assignment $(x, q, p, z) \mapsto (R, T, D, S)$ defines a Poisson endomorphism of $\mathcal{P}_2$ that is not an automorphism. This disproves the Poisson Conjecture for two canonical pairs, and hence for every number of canonical pairs at least two. The associated polynomial map of $\mathbb{A}^4$ preserves the canonical symplectic form, has Jacobian determinant one, and has an explicit fiber consisting of exactly three points. The proof uses a polynomial source coordinate system in which the symplectic identity reduces to three displayed coefficient identities. A separate appendix uses the same four output polynomials and their Hamiltonian duals to construct an explicit nonautomorphic endomorphism of the fourth Weyl algebra.

2020 Mathematics Subject Classification. Primary 17B63; Secondary 14R15, 16S32, 53D05.

Keywords. Poisson Conjecture, canonical Poisson algebra, symplectic-form-preserving polynomial map, Jacobian Conjecture, Dixmier Conjecture, Weyl algebra.

1 Introduction and rank convention

For a positive integer n , write

$$\mathcal{P}_n = \mathbb{C}[x_1, \dots, x_n, p_1, \dots, p_n] \tag{1}$$

with canonical Poisson bracket

$$\{f, g\} = \sum_{i=1}^n \left(\frac{\partial f}{\partial p_i} \frac{\partial g}{\partial x_i} - \frac{\partial f}{\partial x_i} \frac{\partial g}{\partial p_i} \right). \tag{2}$$

Thus $\{p_i, x_j\} = \delta_{ij}$. In this paper, we use *rank* n to mean the index n in (1), equivalently the number of canonical pairs. This differs from the geometric rank of the Poisson tensor, which is $2n$. In particular, the title’s “rank two” means two canonical pairs, four polynomial variables, and a Poisson tensor of rank four. This is also the indexing convention of Adjmagbo and van den Essen, whose canonical Poisson algebra $\mathcal{P}_n$ has $2n$ polynomial generators [1, Section 1].

The assertion $\text{PC}(n)$ is that every Poisson endomorphism of $\mathcal{P}_n$ is an automorphism. Recall that $\text{JC}(3)$, the three-variable Jacobian Conjecture, asserts that a polynomial map $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ with nonzero constant Jacobian determinant is a polynomial automorphism [6]. The present construction begins with an explicit counterexample to this three-variable assertion. The rank-two Poisson endomorphism is obtained by adjoining a polynomial source coordinate and a Hamiltonian correction, and every Poisson identity is then verified directly.

The construction below grew from the three-variable polynomial map

$$\begin{aligned}\Pi(X, Y, W) &= (1 + XY)^3 W + Y^2(1 + XY)(4 + 3XY), \\ \Sigma(X, Y, W) &= Y + 3X(1 + XY)^2 W + 3XY^2(4 + 3XY), \\ \mathcal{R}(X, Y, W) &= 2X - 3X^2 Y - X^3 W.\end{aligned}\tag{3}$$

This map was announced by Alpöge, whose post thanks Akhil Mathew for asking about the problem and Fable for working on it [2]. Buzzard’s contemporaneous account reports that Mathew suggested the problem to Alpöge and that Fable found the counterexample [4]; the Ulam research note similarly credits Mathew for asking the question and Fable for the work leading to the example [8, Introduction]. The Ulam note independently verifies the determinant and the three-point collision and studies the fibers and global geometry [8, Theorem 3.1 and Section 4]. Its Jacobian determinant is -2 , and the three points

$$\left(0, 0, -\frac{1}{4}\right), \quad \left(1, -\frac{3}{2}, \frac{13}{2}\right), \quad \left(-1, \frac{3}{2}, \frac{13}{2}\right)\tag{4}$$

have common image $(-1/4, 0, 0)$. The standard cotangent lift

$$(X, \Xi) \mapsto (F(X), J_F(X)^{-\top} \Xi), \quad F = (\Pi, \Sigma, \mathcal{R}),$$

is polynomial because $\det J_F = -2$, preserves the canonical one-form $\Xi^{\top} dX$, and is noninjective on the zero section. Thus the announced map already gives a Poisson counterexample on three canonical pairs. The purpose of the present construction is to reach two canonical pairs.

The three-dimensional core used here is the determinant-one normalization

$$G = (\mathcal{R}, -\Pi/2, \Sigma).\tag{5}$$

The four-dimensional Poisson endomorphism is obtained from this core by a polynomial change of source coordinates and a Hamiltonian correction. Every identity needed below is proved directly, so the proof does not depend on an external verification of (3).

The main result is the following.

Theorem 1.1 (Main theorem). *There are explicit polynomials $R, T, D, S \in \mathbb{Q}[x, q, p, z]$ such that*

$$\Phi : \mathcal{P}_2 \longrightarrow \mathcal{P}_2, \quad \Phi(x) = R, \quad \Phi(q) = T, \quad \Phi(p) = D, \quad \Phi(z) = S,\tag{6}$$

is a Poisson endomorphism but not an automorphism. Its associated point map

$$\Phi_{\text{pt}} = (R, T, D, S) : \mathbb{A}_{\mathbb{C}}^4 \longrightarrow \mathbb{A}_{\mathbb{C}}^4$$

satisfies

$$\det J\Phi_{\text{pt}} = 1,$$

and the fiber over $(0, 1/8, 0, 0)$ consists exactly of

$$\left(0, 0, \frac{1}{24}, -\frac{1}{8}\right), \quad \left(1, \frac{2}{3}, \frac{247}{96}, -\frac{89}{64}\right), \quad \left(-1, -\frac{2}{3}, \frac{247}{96}, -\frac{89}{64}\right). \quad (7)$$

Consequently, $\text{PC}(2)$ is false.

The formulas have rational coefficients and therefore define the same counterexample over every field of characteristic zero. We work over $\mathbb{C}$ to match the standard formulations of the Poisson and Jacobian conjectures.

2 Poisson endomorphisms and symplectic maps

From now on, order the generators as (x, q, p, z) and write

$$\{f, g\} = f_p g_x - f_x g_p + f_z g_q - f_q g_z. \quad (8)$$

Thus $\{p, x\} = \{z, q\} = 1$, and the corresponding symplectic form is

$$\omega = dx \wedge dp + dq \wedge dz. \quad (9)$$

We record the elementary criterion used below. It is the four-variable case of [1, Theorem 2.1 and Corollary 2.1].

Lemma 2.1 (Symplectic criterion). *Let*

$$F = (F_1, F_2, F_3, F_4) \in \mathbb{C}[x, q, p, z]^4.$$

The substitution $(x, q, p, z) \mapsto (F_1, F_2, F_3, F_4)$ defines a Poisson endomorphism for (8) if and only if

$$dF_1 \wedge dF_3 + dF_2 \wedge dF_4 = dx \wedge dp + dq \wedge dz. \quad (10)$$

Equivalently,

$$\{F_3, F_1\} = 1, \quad \{F_4, F_2\} = 1,$$

and the four mixed brackets vanish. Whenever these conditions hold,

$$\det JF = 1. \quad (11)$$

Proof. Let $X = (x, q, p, z)^\top$ and

$$\Omega_0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}.$$

Then Ω_0 is the matrix of ω , while the matrix of the generator brackets is $\Omega_0^{-1} = -\Omega_0$. Equation (10) is

$$(JF)^\top \Omega_0 (JF) = \Omega_0.$$

It gives the left inverse $\Omega_0^{-1}(JF)^\top \Omega_0$ for JF . Since the coefficient ring is commutative and the matrix is square, JF is invertible. Taking inverses gives

$$(JF)\Omega_0^{-1}(JF)^\top = \Omega_0^{-1},$$

whose entries are exactly the generator-bracket identities. Since the Poisson bracket is a biderivation, preservation on generators is equivalent to preservation on the whole polynomial algebra.

Finally, in the coordinate order (x, q, p, z) ,

$$\omega^2 = -2 dx \wedge dq \wedge dp \wedge dz,$$

whereas the square of the left side of (10) is

$$-2(\det JF) dx \wedge dq \wedge dp \wedge dz.$$

Thus $\det JF = 1$. □

3 The explicit polynomials

Set

$$a = 1 - 3xq, \quad B = 3x^2p + 2az, \quad \beta = B - 9q^2, \quad (12)$$

and then define

$$y = q - \frac{x\beta}{3}, \quad u = xy. \quad (13)$$

The three core polynomials are

$$R = 2x - 3x^2y - x^3\beta, \quad (14)$$

$$S = y + 3x(1 + u)^2\beta + 3xy^2(4 + 3u), \quad (15)$$

$$T = -\frac{1}{2} \left((1 + u)^3\beta + y^2(1 + u)(4 + 3u) \right). \quad (16)$$

The uncorrected fourth coordinate is

$$D_0 = \frac{1 + 3xq}{2} p - 3q^2 z. \quad (17)$$

Define the Hamiltonian correction

$$\begin{aligned} H = & \frac{y^4}{20}(18u^2 + 78u + 125) \\ & + \frac{3\beta y^2}{10}(u^3 + 5u^2 + 10u - 5) \\ & - \frac{\beta^2}{6}(9u + 2) - \frac{x^2\beta^3}{6}, \end{aligned} \quad (18)$$

and put

$$D = D_0 + H. \quad (19)$$

All displayed expressions are polynomials over $\mathbb{Q}$. Their expanded sizes are included for reference.

polynomial	β	y	R	S	T	H	D
nonzero terms	4	5	2	22	47	137	139
total degree	3	4	3	11	15	23	23

The first coordinate simplifies drastically.

Lemma 3.1. *In the original variables (x, q, p, z) ,*

$$R = x(2 - 3xq). \quad (20)$$

Proof. Using $y = q - x\beta/3$,

$$R = 2x - 3x^2 \left(q - \frac{x\beta}{3} \right) - x^3\beta = 2x - 3x^2q.$$

□

We henceforth use Φ for the algebra endomorphism in (6) and $\Phi_{\text{pt}} = (R, T, D, S)$ for its associated point map.

4 A polynomial source coordinate system

The choice of D_0 makes (x, y, β, D_0) a polynomial coordinate system. Temporarily put

$$Q = y + \frac{x\beta}{3}, \quad A = 1 - 3xQ, \quad C = \frac{1 + 3xQ}{2}, \quad M = \beta + 9Q^2. \quad (21)$$

After substituting (12)–(13), one has $Q = q$, $A = a$, and

$$M = 3x^2p + 2Az, \quad D_0 = Cp - 3Q^2z. \quad (22)$$

Proposition 4.1 (Polynomial source coordinates). *The map*

$$\Psi_0 : \mathbb{A}^4 \longrightarrow \mathbb{A}^4, \quad (x, q, p, z) \longmapsto (x, y, \beta, D_0), \quad (23)$$

is a polynomial automorphism. Its inverse is

$$\begin{aligned} q &= Q = y + \frac{x\beta}{3}, \\ p &= 3Q^2(\beta + 9Q^2) + 2(1 - 3xQ)D_0, \\ z &= \frac{1 + 3xQ}{2}(\beta + 9Q^2) - 3x^2D_0. \end{aligned} \quad (24)$$

Moreover,

$$\det J\Psi_0 = -1. \quad (25)$$

The map

$$\Psi : (x, q, p, z) \longmapsto (x, y, \beta, D) \quad (26)$$

is also a polynomial automorphism, with inverse obtained from (24) by replacing D_0 with $D - H(x, y, \beta)$.

Proof. The equations in (22) form the linear system

$$\begin{pmatrix} 3x^2 & 2A \\ C & -3Q^2 \end{pmatrix} \begin{pmatrix} p \\ z \end{pmatrix} = \begin{pmatrix} M \\ D_0 \end{pmatrix}.$$

Its determinant is

$$-9x^2Q^2 - 2AC = -9x^2Q^2 - (1 - 3xQ)(1 + 3xQ) = -1.$$

Consequently,

$$\begin{pmatrix} p \\ z \end{pmatrix} = \begin{pmatrix} 3Q^2 & 2A \\ C & -3x^2 \end{pmatrix} \begin{pmatrix} M \\ D_0 \end{pmatrix},$$

which is precisely (24).

For the determinant, first pass from (x, q, p, z) to (x, q, β, D_0) . At fixed (x, q) , the relevant block is

$$\begin{pmatrix} 3x^2 & 2(1 - 3xq) \\ (1 + 3xq)/2 & -3q^2 \end{pmatrix},$$

whose determinant is -1 . The change $q \mapsto y = q - x\beta/3$ is a triangular shear of determinant one. Hence (25) holds. Finally, $(x, y, \beta, D_0) \mapsto (x, y, \beta, D_0 + H)$ is another triangular shear, with inverse $D_0 = D - H$, proving the assertion for Ψ . $\square$

In particular, x, y, β are algebraically independent, so calculations in $\mathbb{Q}[X, Y, W]$ may be transported to the subalgebra $\mathbb{Q}[x, y, \beta] \subset \mathcal{P}_2$ without ambiguity.

5 The symplectic calculation

We compute ω in the coordinates (x, y, β, D_0) . Continue to use Q, A, C, M from (21). From (24),

$$p = 3Q^2M + 2AD_0, \quad z = CM - 3x^2D_0. \quad (27)$$

The required differentials are

$$dQ = dy + \frac{\beta}{3} dx + \frac{x}{3} d\beta, \quad (28)$$

$$\begin{aligned} dp = & -6D_0Q dx + (6Q(\beta + 18Q^2) - 6D_0x)dQ \\ & + 3Q^2d\beta + 2A dD_0, \end{aligned} \quad (29)$$

$$\begin{aligned} dz = & \left(\frac{3}{2}QM - 6xD_0\right) dx + \left(\frac{3}{2}xM + 18QC\right) dQ \\ & + C d\beta - 3x^2dD_0. \end{aligned} \quad (30)$$

For example,

$$d(3Q^2M) = 3Q^2d\beta + 6Q(\beta + 18Q^2)dQ$$

and

$$d(2AD_0) = 2A dD_0 - 6D_0Q dx - 6D_0x dQ,$$

which gives (29); the formula for dz follows similarly from $dC = \frac{3}{2}(Q dx + x dQ)$ and $dM = d\beta + 18Q dQ$.

Substituting (29) and (30) into $\omega = dx \wedge dp + dQ \wedge dz$ gives

$$\begin{aligned}\omega &= \frac{9}{2}Q(\beta + 21Q^2) dx \wedge dQ + 3Q^2 dx \wedge d\beta + C dQ \wedge d\beta \\ &\quad + 2A dx \wedge dD_0 - 3x^2 dQ \wedge dD_0.\end{aligned}\tag{31}$$

Indeed, the coefficient of $dx \wedge dQ$ is

$$6Q(\beta + 18Q^2) - 6D_0x - \left(\frac{3}{2}QM - 6xD_0\right) = \frac{9}{2}Q(\beta + 21Q^2).$$

By Lemma 3.1,

$$R = x(2 - 3xQ), \quad dR = 2A dx - 3x^2 dQ.\tag{32}$$

Thus the last two terms of (31) equal $dR \wedge dD_0$. Replacing dQ using (28), we obtain

$$\omega = dR \wedge dD_0 + \Theta,\tag{33}$$

where

$$\Theta = A_1 dx \wedge dy + A_2 dx \wedge d\beta + A_3 dy \wedge d\beta\tag{34}$$

with

$$A_1 = \frac{9}{2}Q(\beta + 21Q^2),\tag{35}$$

$$A_2 = 3Q^2 + \frac{\beta(1 + 3xQ)}{6} + \frac{3}{2}xQ(\beta + 21Q^2),\tag{36}$$

$$A_3 = \frac{1 + 3xQ}{2}.\tag{37}$$

The correction H was chosen so that Θ is exactly the remaining part of the target symplectic form.

Proposition 5.1 (Coefficient identity). *For the polynomials R, S, T, H in (14)–(18), regarded as elements of $\mathbb{Q}[x, y, \beta]$,*

$$dR \wedge dH + dT \wedge dS = \Theta.\tag{38}$$

Proof. For $f, g \in \mathbb{Q}[x, y, \beta]$, the coefficients of $df \wedge dg$ are

$$\begin{aligned}[dx \wedge dy](df \wedge dg) &= f_x g_y - f_y g_x, \\ [dx \wedge d\beta](df \wedge dg) &= f_x g_\beta - f_\beta g_x, \\ [dy \wedge d\beta](df \wedge dg) &= f_y g_\beta - f_\beta g_y.\end{aligned}$$

Direct differentiation and collection give

$$[dx \wedge dy](dR \wedge dH + dT \wedge dS) = \frac{1}{2}(x\beta + 3y)(7x^2\beta^2 + 42xy\beta + 3\beta + 63y^2),\tag{39}$$

$$\begin{aligned}[dx \wedge d\beta](dR \wedge dH + dT \wedge dS) &= \frac{1}{6}(7x^4\beta^3 + 63x^3y\beta^2 + 6x^2\beta^2 \\ &\quad + 189x^2y^2\beta + 24xy\beta + \beta + 189xy^3 + 18y^2),\end{aligned}\tag{40}$$

$$[dy \wedge d\beta](dR \wedge dH + dT \wedge dS) = \frac{1}{2}(x^2\beta + 3xy + 1).\tag{41}$$

Appendix A records every derivative used in this collection.

Since $3Q = x\beta + 3y$, the second factor in (39) is

$$7(x\beta + 3y)^2 + 3\beta = 3(\beta + 21Q^2),$$

so (39) is A_1 . Expanding

$$6A_2 = 18Q^2 + \beta(1 + 3xQ) + 9xQ(\beta + 21Q^2)$$

after substituting $3Q = x\beta + 3y$ gives the numerator in (40). Finally,

$$\frac{1}{2}(x^2\beta + 3xy + 1) = \frac{1}{2}(1 + 3xQ) = A_3.$$

The three coefficients agree with (34), proving (38). $\square$

Theorem 5.2 (Exact Poisson identities). *The polynomials in Section 3 satisfy*

$$\begin{aligned} \{D, R\} &= 1, & \{S, T\} &= 1, \\ \{R, S\} &= \{R, T\} = \{D, S\} = \{D, T\} &= 0. \end{aligned} \tag{42}$$

Consequently, Φ is a Poisson endomorphism and

$$\det J\Phi_{\text{pt}} = 1. \tag{43}$$

Proof. By (33), Proposition 5.1, and $D = D_0 + H$,

$$\omega = dR \wedge dD_0 + dR \wedge dH + dT \wedge dS = dR \wedge dD + dT \wedge dS.$$

Apply Lemma 2.1 with $(F_1, F_2, F_3, F_4) = (R, T, D, S)$. $\square$

There is also a short independent check of the first canonical bracket. From (20) and (17),

$$R_x = 2(1 - 3xq), \quad R_q = -3x^2, \quad (D_0)_p = \frac{1 + 3xq}{2}, \quad (D_0)_z = -3q^2.$$

Hence

$$\{D_0, R\} = \frac{1 + 3xq}{2} 2(1 - 3xq) + (-3q^2)(-3x^2) = 1. \tag{44}$$

Proposition 6.2 below shows that R is a Casimir on $\mathbb{C}[x, y, \beta]$; since H lies in that subalgebra, $\{H, R\} = 0$. Thus $\{D, R\} = 1$.

6 The three-dimensional core

This section records the structural relation with (3). Regard R, T, S as polynomials in the independent variables (x, y, β) , and set

$$G = (R, T, S) : \mathbb{A}_{x,y,\beta}^3 \longrightarrow \mathbb{A}^3. \tag{45}$$

Proposition 6.1. *One has*

$$\det \frac{\partial(R, T, S)}{\partial(x, y, \beta)} = 1. \tag{46}$$

Proof. Put

$$w = 1 + xy, \quad \alpha = 2 - 3xy - x^2\beta. \quad (47)$$

Then

$$\det \frac{\partial(x, w, \alpha)}{\partial(x, y, \beta)} = -x^3. \quad (48)$$

In the localization $\mathbb{Q}[x, x^{-1}, w, \alpha]$, define

$$b = 2 + 4w - 3\alpha w^2, \quad c = \frac{1}{2}(\alpha w^3 - w^2 - w). \quad (49)$$

Substitution into (14)–(16) gives

$$R = x\alpha, \quad T = x^{-2}c, \quad S = x^{-1}b. \quad (50)$$

Therefore

$$\det \frac{\partial(R, T, S)}{\partial(x, w, \alpha)} = x^{-3}\Delta, \quad (51)$$

where

$$\Delta = \det \begin{pmatrix} \alpha & 0 & 1 \\ -2c & c_w & c_\alpha \\ -b & b_w & b_\alpha \end{pmatrix}.$$

From (49),

$$b_w = 4 - 6\alpha w, \quad b_\alpha = -3w^2, \quad c_w = \frac{1}{2}(3\alpha w^2 - 2w - 1), \quad c_\alpha = \frac{1}{2}w^3.$$

Expansion along the first row gives

$$\begin{aligned} \Delta &= \alpha(c_w b_\alpha - c_\alpha b_w) + bc_w - 2cb_w \\ &= -\frac{\alpha w^2}{2}(3\alpha w^2 - 2w - 3) \\ &\quad + \frac{1}{2}(3\alpha^2 w^4 - 2\alpha w^3 - 3\alpha w^2 - 2) \\ &= -1. \end{aligned}$$

Combining (48) and (51) proves (46) in the localization. Both sides are polynomials, so the identity holds in $\mathbb{Q}[x, y, \beta]$. $\square$

The induced three-variable Poisson structure makes the role of R transparent.

Proposition 6.2 (Induced Jacobian Poisson bracket). *On $\mathbb{C}[x, y, \beta] \subset \mathcal{P}_2$,*

$$\{x, y\} = x^3, \quad \{x, \beta\} = -3x^2, \quad \{y, \beta\} = -2 + 6xy + 3x^2\beta. \quad (52)$$

For all $f, g \in \mathbb{C}[x, y, \beta]$,

$$\{f, g\} = -\det \frac{\partial(R, f, g)}{\partial(x, y, \beta)}. \quad (53)$$

In particular, R is a Casimir of this three-variable Poisson algebra, and

$$\{S, T\} = 1, \quad \{R, S\} = \{R, T\} = 0.$$

Proof. Since $\beta_p = 3x^2$ and $\beta_z = 2(1 - 3xq)$,

$$\{x, \beta\} = -3x^2, \quad \{q, \beta\} = -2 + 6xq.$$

Using $y = q - x\beta/3$ gives the remaining identities in (52). Moreover,

$$R_x = 2 - 6xy - 3x^2\beta, \quad R_y = -3x^2, \quad R_\beta = -x^3.$$

The bivector encoded by (52) is

$$-R_\beta \partial_x \wedge \partial_y + R_y \partial_x \wedge \partial_\beta - R_x \partial_y \wedge \partial_\beta,$$

which is exactly (53). The claims involving R follow immediately. Finally, Proposition 6.1 gives

$$\{S, T\} = -\det \frac{\partial(R, S, T)}{\partial(x, y, \beta)} = 1.$$

□

The point map factors as

$$\Phi_{\text{pt}} = \sigma \circ (G \times \text{id}) \circ \Psi, \quad \sigma(r, t, s, e) = (r, t, e, s). \quad (54)$$

Indeed, Ψ produces (x, y, β, D) , then $G \times \text{id}$ produces (R, T, S, D) , and σ interchanges the last two coordinates. The determinant factors are

$$\det J\Psi = -1, \quad \det J(G \times \text{id}) = 1, \quad \det J\sigma = -1,$$

which gives a second proof of (43). This factorization also explains the inherited three-point fiber; the two-form calculation in Section 5 is what establishes the full Poisson property.

7 Noninvertibility and the exact three-point fiber

Proposition 7.1. *The Poisson endomorphism Φ is not an algebra automorphism.*

Proof. By Lemma 3.1,

$$\Phi(x) = R = x(2 - 3xq).$$

The two factors are comaximal because

$$1 = \frac{1}{2}(2 - 3xq) + \frac{3q}{2}x. \quad (55)$$

The Chinese remainder theorem therefore gives

$$\begin{aligned} \mathcal{P}_2/(R) &\cong \mathcal{P}_2/(x) \times \mathcal{P}_2/(2 - 3xq) \\ &\cong \mathbb{C}[q, p, z] \times \mathbb{C}[x, x^{-1}, p, z]. \end{aligned} \quad (56)$$

This ring is not a domain, so (R) is not prime. By contrast, $\mathcal{P}_2/(x) \cong \mathbb{C}[q, p, z]$ is a domain. An automorphism would send the prime ideal (x) to the prime ideal $(\Phi(x)) = (R)$, contradicting (56). □

Equivalently, a polynomial-ring automorphism sends an irreducible element to an irreducible element up to a unit, whereas R is a product of two nonunits. The point map also has a completely explicit failure of injectivity.

Proposition 7.2 (Exact fiber). *The fiber of Φ_{pt} over*

$$(R, T, D, S) = \left(0, \frac{1}{8}, 0, 0\right) \quad (57)$$

consists exactly of the three points in (7).

Proof. Because $\Psi = (x, y, \beta, D)$ is a polynomial automorphism and R, T, S depend only on x, y, β , it is enough first to solve

$$R = 0, \quad S = 0, \quad T = \frac{1}{8} \quad (58)$$

in (x, y, β) , with $D = 0$.

If $x = 0$, then (15) gives $S = y$, so $y = 0$; then (16) gives $T = -\beta/2$, whence $\beta = -1/4$.

Suppose $x \neq 0$, and use w, α from (47). The localized formulas (50) read

$$\begin{aligned} R &= x\alpha, \\ S &= x^{-1}(2 + 4w - 3\alpha w^2), \\ T &= \frac{1}{2x^2}(\alpha w^3 - w^2 - w). \end{aligned} \quad (59)$$

The first two equations of (58) give $\alpha = 0$ and $w = -1/2$. The third formula then becomes

$$T = \frac{1}{8x^2}.$$

Since $T = 1/8$, one has $x^2 = 1$. Moreover,

$$xy = w - 1 = -\frac{3}{2}, \quad x^2\beta = 2 - 3xy = \frac{13}{2}.$$

Thus the complete fiber in (x, y, β, D) -coordinates is

$$\left(0, 0, -\frac{1}{4}, 0\right), \quad \left(1, -\frac{3}{2}, \frac{13}{2}, 0\right), \quad \left(-1, \frac{3}{2}, \frac{13}{2}, 0\right). \quad (60)$$

It remains to recover (q, p, z) . Direct substitution in (18) gives

$$H\left(0, 0, -\frac{1}{4}\right) = -\frac{1}{48}, \quad H\left(\pm 1, \mp \frac{3}{2}, \frac{13}{2}\right) = -\frac{1097}{192}. \quad (61)$$

Since $D = 0$, the coordinate $D_0 = D - H$ equals $1/48$ at the first point and $1097/192$ at the other two. At $(x, y, \beta) = (0, 0, -1/4)$, the inverse formulas (24) give

$$q = 0, \quad p = \frac{1}{24}, \quad z = -\frac{1}{8}.$$

At $(x, y, \beta) = (1, -3/2, 13/2)$, one has

$$Q = \frac{2}{3}, \quad A = -1, \quad C = \frac{3}{2}, \quad M = \frac{21}{2},$$

and hence

$$p = 3 \left(\frac{2}{3} \right)^2 \frac{21}{2} - 2 \left(\frac{1097}{192} \right) = \frac{247}{96},$$

$$z = \frac{3}{2} \frac{21}{2} - 3 \left(\frac{1097}{192} \right) = -\frac{89}{64}.$$

The case $(-1, 3/2, 13/2)$ gives $Q = -2/3$ and the same values of p, z . These are exactly the points in (7). $\square$

Remark 7.3 (Gröbner-basis certificate). Let

$$I = \left(R, S, T - \frac{1}{8} \right) \subset \mathbb{Q}[x, y, \beta],$$

where R, S, T are regarded as polynomials in the independent variables (x, y, β) . With lexicographic order $\beta > y > x$, exact reduction gives the reduced Gröbner basis

$$\beta - \frac{27x^2 - 1}{4}, \quad y + \frac{3x}{2}, \quad x(x-1)(x+1). \quad (62)$$

This independently certifies that the core fiber is exactly the three points in (60). Since the final polynomial in (62) is squarefree in characteristic zero, the fiber scheme is reduced.

Theorem 1.1 now follows from Theorem 5.2, Proposition 7.1, and Proposition 7.2.

8 Extension to higher Poisson rank

Corollary 8.1. *The Poisson Conjecture $\text{PC}(n)$ is false for every $n \geq 2$.*

Proof. For $n > 2$, extend Φ by the identity on the remaining canonical pairs. The extended map is Poisson, while the irreducible generator x still has the reducible image $x(2 - 3xq)$; hence the extension cannot be an automorphism. $\square$

Appendix B gives a separate explicit Hamiltonian construction in the fourth Weyl algebra. It is not used in the proof of the Poisson theorem: the appendix directly verifies the Weyl relations and proves non-surjectivity. Its purpose is the explicit Hamiltonian witness attached to the rank-two Poisson map, rather than merely the existence of a counterexample at that Weyl index.

9 Exact computational audit

The proof above is algebraic and does not depend on computer algebra. Two exact programs accompany this source as independent transcription and expansion checks.

The first, `verify_rank2_poisson_sympy.py`, uses SymPy over $\mathbb{Q}$ [7]. It verifies the factorization (20), all six brackets in (42), the source inverse (24), all three coefficients in (38), the core determinant (46), the induced brackets (52), the localized formulas (50), the three-point fiber, the Gröbner basis (62), and the size data in Section 3. It verifies the four-variable determinant through the factorization (54), rather than by a needlessly large direct determinant expansion. For Appendix B, it assembles the full matrix $(\delta_i(f_j))$ from the verified brackets and checks that

all pairwise brackets of the four Hamiltonians are constant; these are precisely the identities behind $\mathbf{AJ}_f^\top = I_4$ and the commuting derivations.

The second, `verify_rank2_poisson_sparse.py`, implements sparse multivariate polynomials and exact rational arithmetic using only Python's standard library. It imports no computer algebra system and independently checks the six brackets, source inverse, coefficient identity, core determinant, collision, and size data. Both programs return zero residuals for every identity encoded in their respective audit suites.

A Derivative certificate for the coefficient identity

This appendix supplies the derivative list behind Proposition 5.1. Put

$$w = 1 + u.$$

Define

$$P_1(u) = 18u^2 + 78u + 125, \quad P_2(u) = u^3 + 5u^2 + 10u - 5.$$

Then

$$P'_1(u) = 36u + 78, \quad P'_2(u) = 3u^2 + 10u + 10.$$

Treating x, y, β as independent variables, differentiation gives

$$R_x = 2 - 6u - 3x^2\beta, \quad R_y = -3x^2, \quad R_\beta = -x^3, \quad (63)$$

$$S_x = 3\beta(1+u)(1+3u) + 6y^2(2+3u), \quad (64)$$

$$S_y = 1 + 6x^2\beta(1+u) + 24u + 27u^2, \quad S_\beta = 3x(1+u)^2, \quad (65)$$

$$T_x = -\frac{1}{2}(3\beta y(1+u)^2 + y^3(7+6u)), \quad (66)$$

$$T_y = -\frac{1}{2}(3\beta x(1+u)^2 + 2y(1+u)(4+3u) + xy^2(7+6u)), \quad (67)$$

$$T_\beta = -\frac{1}{2}(1+u)^3, \quad (68)$$

$$H_x = \frac{y^5}{20}P'_1(u) + \frac{3\beta y^3}{10}P'_2(u) - \frac{3}{2}\beta^2 y - \frac{1}{3}x\beta^3, \quad (69)$$

$$H_y = \frac{4y^3 P_1(u) + xy^4 P'_1(u)}{20} + \frac{3\beta}{10}(2yP_2(u) + xy^2 P'_2(u)) - \frac{3}{2}x\beta^2, \quad (70)$$

$$H_\beta = \frac{3y^2}{10}P_2(u) - \frac{\beta(9u+2)}{3} - \frac{1}{2}x^2\beta^2. \quad (71)$$

Substituting (63)–(71) into

$$\begin{aligned} C_{xy} &= R_x H_y - R_y H_x + T_x S_y - T_y S_x, \\ C_{x\beta} &= R_x H_\beta - R_\beta H_x + T_x S_\beta - T_\beta S_x, \\ C_{y\beta} &= R_y H_\beta - R_\beta H_y + T_y S_\beta - T_\beta S_y \end{aligned}$$

and collecting terms gives

$$\begin{aligned} C_{xy} &= \frac{1}{2}(x\beta + 3y)(7x^2\beta^2 + 42xy\beta + 3\beta + 63y^2), \\ C_{x\beta} &= \frac{1}{6}(7x^4\beta^3 + 63x^3y\beta^2 + 6x^2\beta^2 + 189x^2y^2\beta \\ &\quad + 24xy\beta + \beta + 189xy^3 + 18y^2), \\ C_{y\beta} &= \frac{1}{2}(x^2\beta + 3xy + 1). \end{aligned}$$

These are exactly (39)–(41). The reductions to A_1, A_2, A_3 are carried out in the proof of Proposition 5.1.

B An explicit nonautomorphic endomorphism of the fourth Weyl algebra

Let

$$\mathcal{B} = \mathbb{C}[X_1, X_2, X_3, X_4]$$

and identify (X_1, X_2, X_3, X_4) with (x, q, p, z) . In this appendix, the subscript four counts the four commuting coordinate generators; this is distinct from the canonical-pair convention used for the phrase “rank two” in the title. The fourth Weyl algebra is

$$\mathcal{A}_4 = \mathbb{C}\langle X_1, \dots, X_4, \partial_1, \dots, \partial_4 \rangle,$$

with relations

$$[\partial_i, X_j] = \delta_{ij}, \quad [X_i, X_j] = [\partial_i, \partial_j] = 0.$$

It acts faithfully as the ring of polynomial differential operators on $\mathcal{B}$. The conjecture DC(4) asserts that every endomorphism of $\mathcal{A}_4$ is an automorphism; this is the fourth-rank instance of Dixmier’s problem [5].

The existence of a nonautomorphic endomorphism of $\mathcal{A}_4$ can also be obtained from the announced three-variable Keller map through the standard Jacobian-to-Weyl lift and extension by the identity; see [3, equations (1)–(2) and Proposition 5(3)]. The purpose here is more explicit: we construct a Hamiltonian endomorphism canonically attached to the rank-two Poisson map and prove its non-surjectivity directly.

For $F \in \mathcal{B}$, let

$$H_F = \{F, \cdot\} = F_p \partial_x + F_z \partial_q - F_x \partial_p - F_q \partial_z. \quad (72)$$

Set

$$(f_1, f_2, f_3, f_4) = (R, T, D, S) \quad (73)$$

and

$$\delta_1 = H_D, \quad \delta_2 = H_S, \quad \delta_3 = -H_R, \quad \delta_4 = -H_T. \quad (74)$$

Each f_i acts by multiplication, and each δ_i is an explicit first-order differential operator with polynomial coefficients.

Theorem B.1. *The assignment*

$$\widehat{\Phi} : \mathcal{A}_4 \longrightarrow \mathcal{A}_4, \quad X_i \longmapsto f_i, \quad \partial_i \longmapsto \delta_i \quad (1 \leq i \leq 4), \quad (75)$$

defines an algebra endomorphism that is not an automorphism. Consequently, DC(4) is false.

Proof. The six identities in (42) give

$$\delta_i(f_j) = \delta_{ij}. \quad (76)$$

For example, $\delta_1(R) = \{D, R\} = 1$ and $\delta_3(D) = -\{R, D\} = 1$. Therefore

$$[\delta_i, f_j] = \delta_{ij}$$

as differential operators on $\mathcal{B}$. The multiplication operators f_i commute. Hamiltonian derivations satisfy

$$[H_F, H_G] = H_{\{F, G\}}, \quad (77)$$

and every Poisson bracket between the Hamiltonians in (74) is constant. Hence $[\delta_i, \delta_j] = 0$. The images in (75) satisfy all defining Weyl relations, so $\widehat{\Phi}$ is an endomorphism. This Hamiltonian lift is the present specialization of the construction in the proof of [3, Theorem 7(2), equation (5)]; all relations needed here have just been verified directly.

It remains to prove non-surjectivity. Filter $\mathcal{A}_4$ by differential order and write

$$\delta_i = \sum_{k=1}^4 a_{ik} \partial_k, \quad \mathbf{A} = (a_{ik}). \quad (78)$$

Let J_f be the Jacobian matrix of (f_1, f_2, f_3, f_4) with respect to (X_1, X_2, X_3, X_4) . Equation (76) is the matrix identity

$$\mathbf{A} J_f^\top = I_4. \quad (79)$$

Since $\det J_f = 1$, the matrix $\mathbf{A}$ is invertible over $\mathcal{B}$. The same Jacobian identity also implies that the f_i are algebraically independent. Indeed, a nonzero relation of minimal total degree would, after differentiation and multiplication by J_f^{-1} , give lower-degree relations from all its first derivatives; minimality would force those derivatives to be zero, so the original relation would be constant. For the order filtration, let σ_m denote the degree- m principal-symbol map, and put

$$\eta_i = \sigma_1(\delta_i), \quad \xi = (\xi_1, \dots, \xi_4)^\top, \quad \eta = (\eta_1, \dots, \eta_4)^\top.$$

In the associated graded algebra

$$\text{gr } \mathcal{A}_4 \cong \mathcal{B}[\xi_1, \dots, \xi_4], \quad (80)$$

one has $\eta = \mathbf{A}\xi$. Since $\mathbf{A} \in \text{GL}_4(\mathcal{B})$,

$$\mathcal{B}[\eta_1, \dots, \eta_4] = \mathcal{B}[\xi_1, \dots, \xi_4].$$

In particular, the monomials η^β are linearly independent over $\mathcal{B}$, and

$$\sigma_{|\beta|}(f^\alpha \delta^\beta) = f^\alpha \eta^\beta. \quad (81)$$

Assume for contradiction that $\widehat{\Phi}$ is an automorphism. Applying it to the standard Poincaré–Birkhoff–Witt basis shows that

$$\{f^\alpha \delta^\beta : \alpha, \beta \in \mathbb{N}^4\} \quad (82)$$

is a $\mathbb{C}$ -basis of $\mathcal{A}_4$. Let $g \in \mathcal{B}$ be a multiplication operator and write its unique finite expansion as

$$g = \sum_{\alpha, \beta} c_{\alpha\beta} f^\alpha \delta^\beta.$$

Suppose

$$m = \max\{|\beta| : c_{\alpha\beta} \neq 0 \text{ for some } \alpha\}$$

is positive. Since g has differential order zero, the order- m principal symbol must vanish:

$$\sum_{|\beta|=m} \left(\sum_{\alpha} c_{\alpha\beta} f^{\alpha} \right) \eta^{\beta} = 0. \quad (83)$$

By the linear independence established above, each coefficient $\sum_{\alpha} c_{\alpha\beta} f^{\alpha}$ is zero. By the algebraic independence just noted, every scalar $c_{\alpha\beta}$ with $|\beta| = m$ is zero, a contradiction. Therefore every multiplication operator belongs to $\mathbb{C}[f_1, f_2, f_3, f_4]$, and

$$\mathcal{B} = \mathbb{C}[f_1, f_2, f_3, f_4]. \quad (84)$$

Since the f_i are algebraically independent, the polynomial endomorphism $(X_1, X_2, X_3, X_4) \mapsto (f_1, f_2, f_3, f_4)$ is injective; equation (84) makes it surjective. It would therefore be an automorphism, contrary to Proposition 7.1. Thus $\hat{\Phi}$ is not an automorphism. $\square$

Remark B.2 (Why the fourth Weyl algebra appears). The main construction has four algebraically independent output coordinates f_1, f_2, f_3, f_4 . Pairing them with the four Hamiltonian derivations in (74) produces four multiplication–derivation pairs, which is precisely the generator count for $\mathcal{A}_4$. Thus the same polynomials naturally connect the rank-two canonical Poisson algebra—where “rank two” counts two canonical pairs—with the fourth Weyl algebra, whose index counts four polynomial coordinates and their four conjugate derivations.

AI provenance, use, and author responsibility

The three-variable core used here was announced by Levent Alpöge. As documented in Section 1, the announcement and contemporaneous accounts credit Akhil Mathew with raising the problem and Fable with the work leading to the counterexample. The four-variable construction, including the Hamiltonian correction H , was produced during an interactive research session with ChatGPT 5.6 Sol. ChatGPT 5.6 Sol also assisted with the differential-form organization of the proof, exact symbolic verification, literature checking, and drafting of this manuscript. Claude Fable 5 subsequently supplied independent algebraic audits and editorial comments. The human author bears full responsibility for checking the mathematics, exposition, source attribution, and every claim in this manuscript.

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